

Ques. WHAT IS THE DIFFERENCE BETWEEN HIGH-POWER LEDs AND LOW-POWER LEDs?**Pamarthi Kanakaraja****Ans.** The differences are as follows:

Light output. The most important difference between low- and high-power LEDs is the amount of light each puts out. Low-power ones emit a few lumens at most—usually in single figures—although newer white or blue ones can be slightly brighter. A typical 3W high-power LED, on the other hand, has a light flux of over 150 lumens and some very high-end ones can exceed 1000 lumens.

Heat sensitivity. Heat sensitivity is not a problem with low-power LEDs, but modern high-power ones use a much higher current. Hundreds of milliamps is normal; some go over 1A. Even with their high efficiency, a lot more heat is produced, and unless properly managed, it is likely that the LEDs could fail pretty quickly—usually within seconds.

High-power LEDs will always be mounted on some kind of heat-sink for this reason. Often it is simply the casing of the device these are in, like the body of a flashlight. LED light bulbs usually have a large heat-sink built into the base.

Power consumption. A 1W to 3W LED is usually used for high-power LED lights. Low-power LED lights adopt SMD LEDs, which mainly include SMD 5050 0.2W each, SMD 3258 0.06W each, SMD 3014 0.01W each, SMD 5630, 0.5W each and SMD 5730 0.5W each.

Cost. The cost of high-power LEDs is higher than of low-power LEDs. First, the cost itself is high. Second, high-power LEDs require an aluminium heat-sink, while low-power ones only need an ordinary circuit board. Low-power LED could also dissipate heat naturally.

Application. High-power LEDs are used for LED ceiling lights, LED bulbs, LED streetlights and so on. Whereas, low-power LEDs are used for LED panel lights, LED downlights, LED tubelights, LED strips and so on.

Q2. WHAT IS VLSI DESIGN FLOW, Y-CHART AND DESIGN HIERARCHY IN VLSI SYSTEMS?**Sachin Soni**

A2. The VLSI design process at various levels is usually evolutionary in nature. First, the initial design is prepared and tested against all requirements.

Y-chart, first introduced by D. Gajski, represents a complete VLSI design flow used for most logic chips using design activities on three different axes that resemble the letter Y.

Y-chart mainly consists of three domains: behavioural, structural and geometrical layout. Each domain can be divided into levels of abstraction, using concentric rings. At the top level (outer ring), we consider the architecture of the chip, and at lower levels (inner rings) we refine the design into detailed implementation.

In VLSI design flow, various abstractions of design are behavioural, logic, circuit and mask layout. Here, verification of the design plays a very important role in every step during the design process as given below.

System specification. Specifications always come first. At this stage, designing specifications as per the market requirement is done; these describe the functionality, interface and architecture of the digital IC to be designed.

Architecture. With the help of the specification sheet, the target IC's architecture is decided and a layout for the same is designed using EDA tools. In the next step, this architecture is

implemented and tested with the help of hardware description language and its tools.

RTL coding. RTL is an acronym for register-transfer level. It means VHDL/Verilog code written based on the architecture, how data is transformed and how it is passed from register to register.

RTL verification. In VLSI design, RTL simulation and verification is an important step, which ensures that your design is logically correct and without timing errors. Various simulation tools like ModelSim or synopsys tools can be used for verification.

Synthesis. At this stage the design gets physical, which means that RTL is converted into some design in terms of logic gates, which drive the circuit and architecture.

Back-end. After synthesis, the final design is handed over to the IC manufacturer.

Tape-out and foundry. This step is part of the backend where the final result of first five steps, that is frontend, is provided to the manufacturer in the form of a photomask. Then, the manufacturer performs wafer processing, testing, packaging, etc. to test the physical IC. Finally, when all requirements are met, the design is sent for mass production.

The design hierarchy in VLSI is based on divide-and-conquer technique. In this, the task is divided into smaller tasks so that the design reaches the simplest level and manufacturing becomes easier. You may refer following link for details: www.tutorialspoint.com/vlsi_design_digital_system.htm

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